

Health Misinformation Survey Analysis

This notebook explores how social media behavior relates to trust in health information among student respondents.

We will:

- clean and prepare survey data
- summarize key demographic and behavioral patterns
- visualize trust distributions and relationships
- extract quick, decision-friendly insights

```
In [10]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from pathlib import Path

sns.set_theme(style="whitegrid", palette="Set2")
plt.rcParams["figure.figsize"] = (10, 6)

pd.set_option("display.max_columns", None)
pd.set_option("display.width", 140)
```

1. Load Data

The code below checks common file locations and loads the first match.

```
In [11]: candidate_paths = [
    Path("HealthMisinformation.csv"),
    Path(r"c:/Users/josep/OneDrive/Documents/GW 2026/Foundations of Research PUBH 2140/HealthMisinformation.csv")
]

csv_path = next((p for p in candidate_paths if p.exists()), None)
if csv_path is None:
    raise FileNotFoundError("Could not find HealthMisinformation.csv. Place it in the correct location.")

df_raw = pd.read_csv(csv_path)
print(f"Loaded file: {csv_path}")
print(f"Rows: {df_raw.shape[0]}, Columns: {df_raw.shape[1]}")
df_raw.head()
```

```
Loaded file: c:\Users\josep\OneDrive\Documents\GW 2026\Foundations of Research PUBH 2140\HealthMisinformation.csv
Rows: 12, Columns: 41
```

Out[11]:

	Timestamp	Total score	Do you consent to participate?	Do you consent to participate? [Score]	Do you consent to participate? [Feedback]	What is your age range?	What is your gender?
0	2026/03/26 12:36:20 PM AST	0.00 / 0	Yes - Continue to survey	-- / 0	NaN	24+	
1	2026/03/26 12:48:35 PM AST	0.00 / 0	Yes - Continue to survey	-- / 0	NaN	18-20	
2	2026/03/26 12:50:41 PM AST	0.00 / 0	Yes - Continue to survey	-- / 0	NaN	18-20	
3	2026/03/26 1:36:57 PM AST	0.00 / 0	Yes - Continue to survey	-- / 0	NaN	18-20	
4	2026/03/26 1:37:34 PM AST	0.00 / 0	Yes - Continue to survey	-- / 0	NaN	21-23	

2. Prepare Analysis Dataset

The survey export includes score/feedback helper columns. Here we retain core analytical variables and convert trust responses to numeric Likert values (1-5).

```
In [12]: df = df_raw.copy()

selected_cols = {
    "What is your age range?": "age_range",
    "What is your gender?": "gender",
    "What is your academic year?": "academic_year",
    "How often do you use social media?": "social_media_use",
    "Which platforms do you use most frequently?": "platforms",
    "How often do you encounter health-related content on social media?": "health_c",
    "I trust doctors and healthcare professionals.": "trust_doctors",
    "I trust public health agencies (e.g., CDC, WHO.)": "trust_public_health",
    "I trust hospitals and healthcare systems. ": "trust_hospitals",
    "I trust government leadership in making health-related decisions.": "trust_gov",
    "Information from health professionals is more reliable than social media. ": "
```

```

    "In your own words, how does social media influence your trust in health inform
}

missing = [c for c in selected_cols if c not in df.columns]
if missing:
    raise KeyError(f"Missing expected columns: {missing}")

analysis = df[list(selected_cols.keys())].rename(columns=selected_cols)

trust_cols = [
    "trust_doctors",
    "trust_public_health",
    "trust_hospitals",
    "trust_government",
    "professionals_more_reliable"
]

for c in trust_cols:
    analysis[c] = pd.to_numeric(analysis[c], errors="coerce")

analysis["avg_trust"] = analysis[trust_cols].mean(axis=1)
analysis.head()

```

Out[12]:

	age_range	gender	academic_year	social_media_use	platforms	health_co
--	-----------	--------	---------------	------------------	-----------	-----------

0	24+	Male	Senior	1-2 hours	Tiktok;YouTube;Facebook	
1	18-20	Male	Freshman	1-2 hours	Tiktok;Instagram;YouTube	
2	18-20	Male	Sophomore	3-4 hours	Tiktok;Instagram;YouTube	
3	18-20	Female	Sophomore	Less than 1 hour per day	Tiktok	
4	21-23	Male	Junior	3-4 hours	Tiktok;Instagram	

3. Quick Profile of Respondents

This view shows who responded and how much they use social media.

```

In [13]: fig, axes = plt.subplots(1, 2, figsize=(14, 5))

age_order = analysis["age_range"].value_counts().index
sns.countplot(data=analysis, x="age_range", order=age_order, ax=axes[0], color="#4C
axes[0].set_title("Age Range Distribution")
axes[0].set_xlabel("Age Range")
axes[0].set_ylabel("Count")

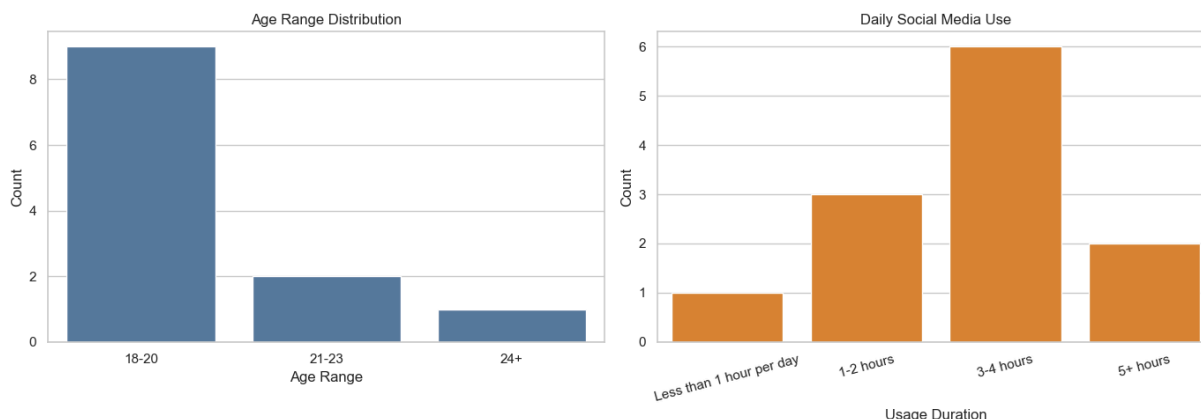
```

```

use_order = ["Less than 1 hour per day", "1-2 hours", "3-4 hours", "5+ hours"]
sns.countplot(data=analysis, x="social_media_use", order=use_order, ax=axes[1], col
axes[1].set_title("Daily Social Media Use")
axes[1].set_xlabel("Usage Duration")
axes[1].set_ylabel("Count")
axes[1].tick_params(axis="x", rotation=15)

plt.tight_layout()
plt.show()

```



Insight

A quick profile often reveals whether interpretations should be treated as representative or exploratory. In small samples, strong trends are useful but should be validated with larger follow-up data.

4. Platform Popularity (Multi-Select Responses)

Because platform choices are stored as semicolon-separated selections, we split and count each platform mention.

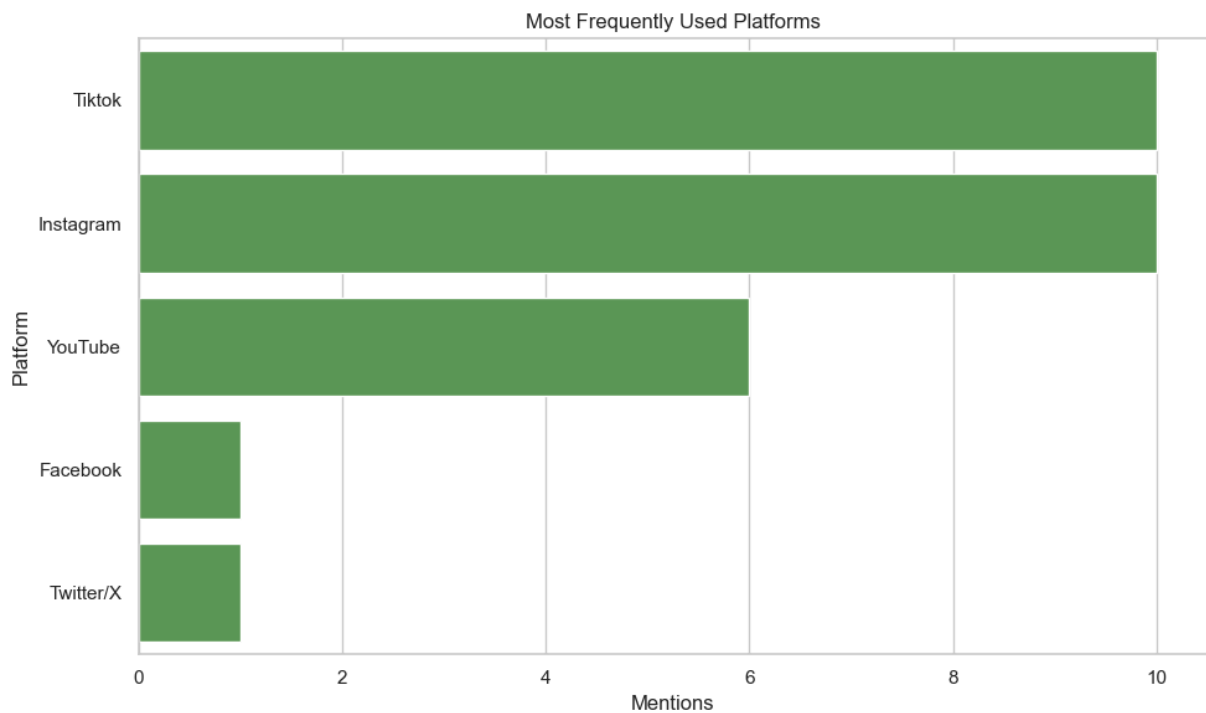
```

In [14]: platform_counts = (
    analysis["platforms"]
    .dropna()
    .str.split(";")
    .explode()
    .str.strip()
    .value_counts()
)

sns.barplot(x=platform_counts.values, y=platform_counts.index, color="#54A24B")
plt.title("Most Frequently Used Platforms")
plt.xlabel("Mentions")
plt.ylabel("Platform")
plt.tight_layout()
plt.show()

platform_counts

```



```
Out[14]: platforms
Tiktok      10
Instagram   10
YouTube      6
Facebook     1
Twitter/X    1
Name: count, dtype: int64
```

Insight

Platform concentration matters for intervention design. If most users cluster on one or two channels, targeted health communication campaigns can be more efficient than broad multi-platform messaging.

5. Trust by Source Type

This chart compares average trust scores across institutions and information sources.

```
In [15]: trust_means = analysis[trust_cols].mean().sort_values(ascending=False)

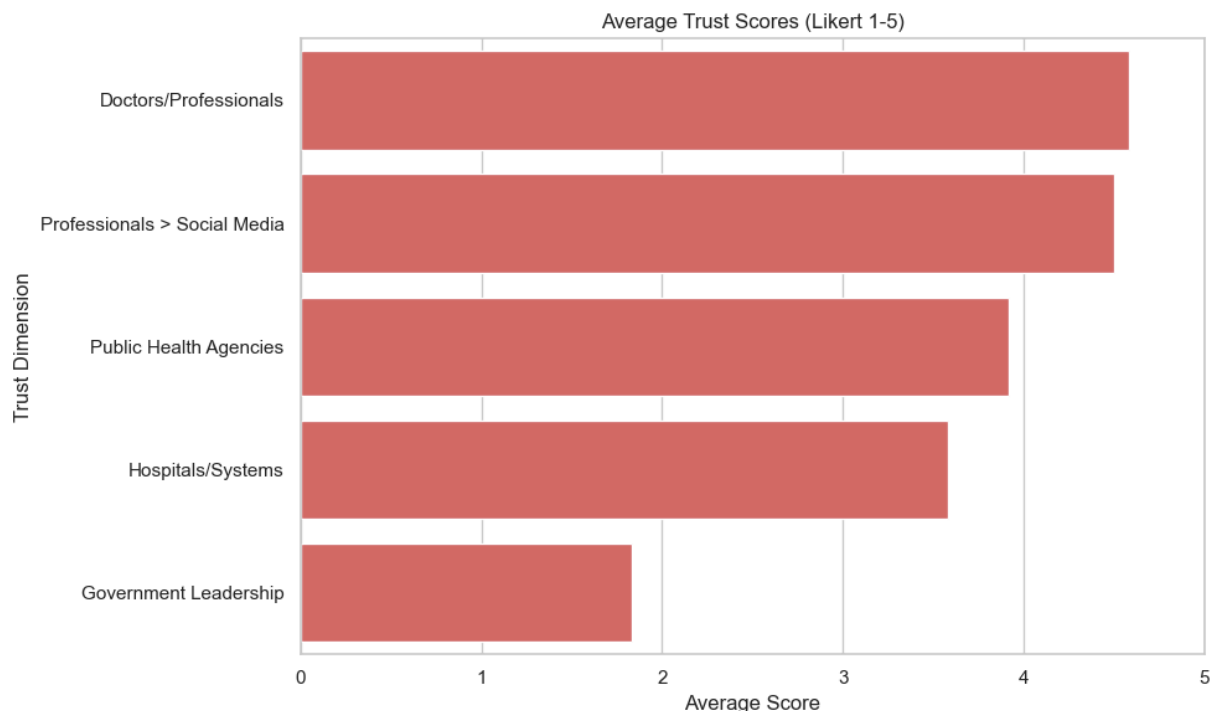
label_map = {
    "trust_doctors": "Doctors/Professionals",
    "trust_public_health": "Public Health Agencies",
    "trust_hospitals": "Hospitals/Systems",
    "trust_government": "Government Leadership",
    "professionals_more_reliable": "Professionals > Social Media"
}

plot_data = trust_means.rename(index=label_map)

sns.barplot(x=plot_data.values, y=plot_data.index, color="#E45756")
```

```
plt.title("Average Trust Scores (Likert 1-5)")
plt.xlabel("Average Score")
plt.ylabel("Trust Dimension")
plt.xlim(0, 5)
plt.tight_layout()
plt.show()
```

plot_data



```
Out[15]: Doctors/Professionals      4.583333
Professionals > Social Media      4.500000
Public Health Agencies           3.916667
Hospitals/Systems                3.583333
Government Leadership            1.833333
dtype: float64
```

Insight

A common pattern in misinformation contexts is lower trust in government than in clinicians or formal healthcare institutions. This can guide who should be the messenger in public health campaigns.

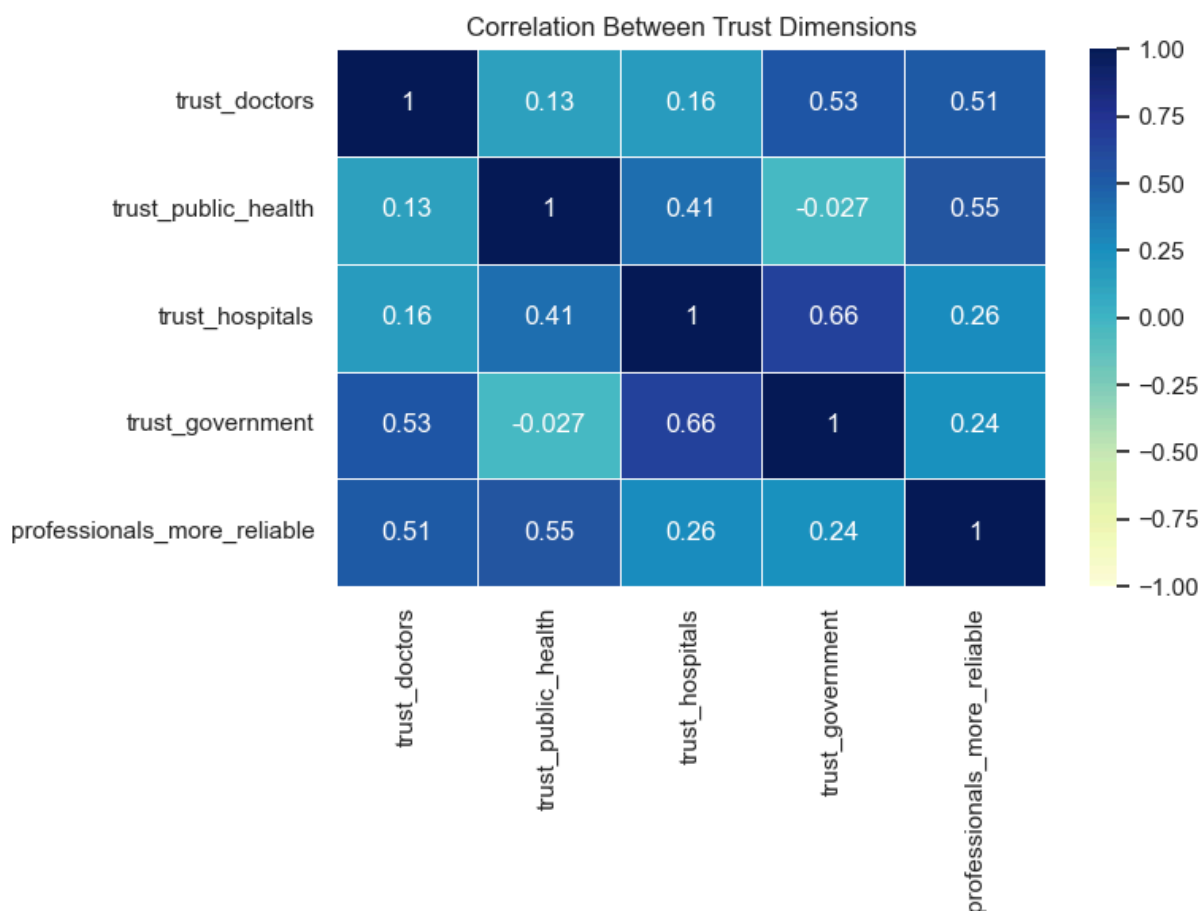
6. Correlation Heatmap of Trust Variables

Correlation helps show whether trust dimensions move together (e.g., trusting doctors and agencies simultaneously).

```
In [16]: corr = analysis[trust_cols].corr()

plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap="YlGnBu", vmin=-1, vmax=1, linewidths=0.5)
```

```
plt.title("Correlation Between Trust Dimensions")
plt.tight_layout()
plt.show()
```



Insight

Higher positive correlations suggest that trust can function as a broader attitude rather than isolated beliefs. If one trust domain is improved, others may improve with it.

7. Exposure vs Overall Trust

This visualization checks whether seeing health-related content more frequently is associated with higher or lower overall trust.

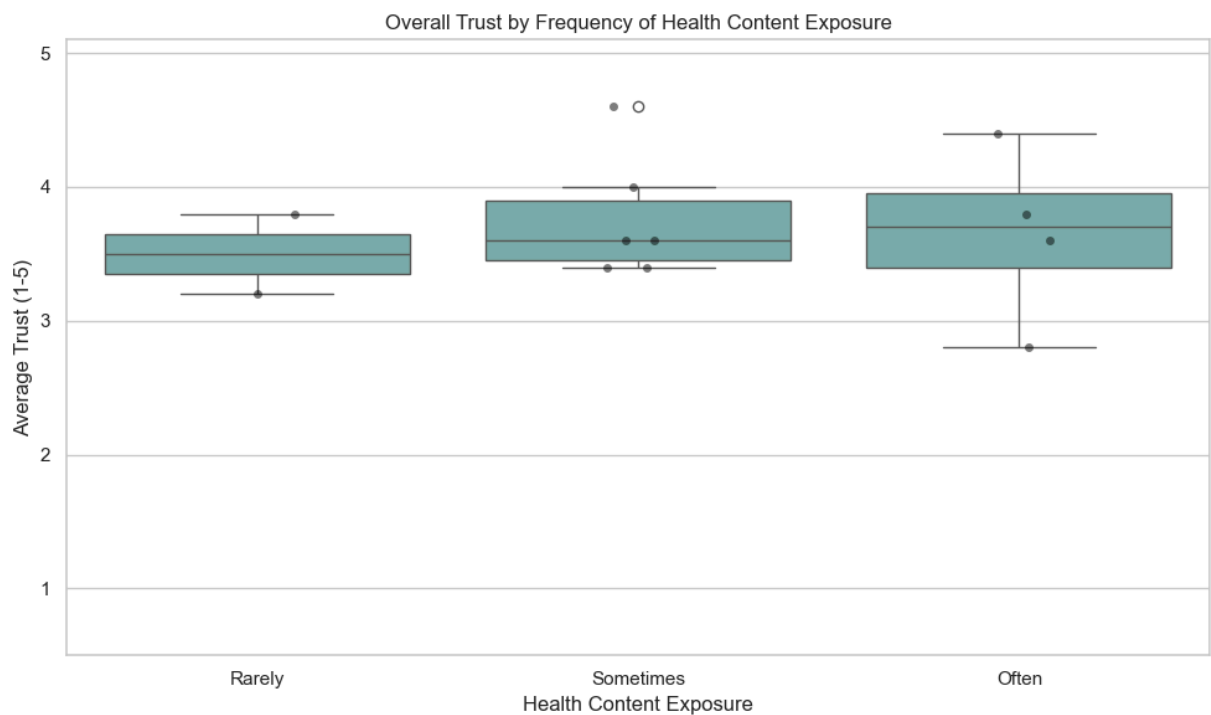
```
In [17]: exposure_order = ["Rarely", "Sometimes", "Often"]

sns.boxplot(
    data=analysis,
    x="health_content_frequency",
    y="avg_trust",
    order=exposure_order,
    color="#72B7B2"
)
sns.stripplot(
```

```
data=analysis,
x="health_content_frequency",
y="avg_trust",
order=exposure_order,
color="black",
alpha=0.5,
size=5
)

plt.title("Overall Trust by Frequency of Health Content Exposure")
plt.xlabel("Health Content Exposure")
plt.ylabel("Average Trust (1-5)")
plt.ylim(0.5, 5.1)
plt.tight_layout()
plt.show()

analysis.groupby("health_content_frequency")["avg_trust"].agg(["count", "mean", "std
```



Out[17]:

	count	mean	std
health_content_frequency			
Often	4	3.650000	0.660808
Rarely	2	3.500000	0.424264
Sometimes	6	3.766667	0.463321

8. Open-Response Snapshot

Qualitative responses help explain *why* numeric trust patterns appear.


```
In [18]: responses = analysis["open_response"].dropna().str.strip()
responses = responses[responses != ""]

print(f"Non-empty responses: {len(responses)}")
responses.head(10)
```

Non-empty responses: 6

```
Out[18]: 1    Social Media give a microphone to any famous p...
        3    It doesn't. I always find myself very skeptica...
        7    It often sells me things that influence how I ...
        8    I find that social media tend to spread misinf...
        9    I don't usually trust the health information t...
        11                                         sometimes
        Name: open_response, dtype: str
```

9. Key Takeaways

1. **Trust is not uniform across institutions;** differences in messenger credibility are visible even in a small sample.
2. **Platform concentration provides intervention leverage;** focusing on dominant channels can improve campaign efficiency.
3. **Health-content exposure may shape trust,** but sample size is limited, so this is best treated as exploratory evidence.
4. **Mixed-method interpretation is strongest:** pair quantitative scores with open responses to understand mechanisms behind trust and skepticism.